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Developing a Clastic Field with Apparent Tilted Contact Using the Stepping Contact Concept in Stratigraphically Trapped Reservoirs

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Understanding fluid contact behavior in clastic reservoirs is crucial for accurately estimating hydrocarbons in place and optimizing field development plans. While most field development strategies rely on assuming horizontal contacts, certain geological settings, particularly those affected by late-stage tectonics, may display apparent tilted contacts. This paper explores a case study in which a clastic field shows such behavior due to a post-charge structural uplift, affecting equilibrium fluid distribution across two distinct reservoirs. The work focuses on evaluating the tilted contact concept and assessing its implications on reservoir development, contrasting it with a simplified stepping contact approach commonly used in current workflows.

This study aims to Evaluate the observed tilted contact using a combination of geological and reservoir engineering data. It compares the current stepping contact mapping approach with the tilted contact concept. It Quantify the time required for capillary equilibrium in both reservoirs, using drainage and imbibition curve analysis. It assesses the implications of contact behavior on field development strategies and potential hydrocarbon volumes.

The scope of this investigation includes integration of well data, structural interpretation, charge history, saturation profiles, and capillary pressure modeling to explain observed field behaviors and provide actionable recommendations for development planning.

Introduction

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Geological Setting and Field Background

The field under study spans approximately 15 km by 8 km and contains over 100 drilled wells. It includes two main reservoirs with different rock qualities:

- The upper reservoir exhibits high permeability and porosity, typically associated with structural traps and conventional hydrocarbon entrapment mechanisms.
- The lower reservoir is characterized by lower rock quality, greater heterogeneity, and a more regional distribution. It appears to be controlled by stratigraphic trapping mechanisms.

Oil-water contacts were mapped using oil-down-to (ODT) and water-up-to (WUT) observations. A distinct tilt in the contact from north to south was observed, suggesting a deviation from horizontal equilibrium. This tilt aligns with a nose-shaped structural feature, indicating post-charge tectonic uplift.

Field X is recognized as a distinct entity from its neighboring field and comprises multiple stacked reservoirs with hydrocarbon potential in two primary zones: Zone 1 and Zone 2. These intervals belong to the Gharif Formation within the Haushi Group and are of Early to Middle Permian age. The combined gross thickness of Zone 1 and Zone 2 is approximately 185 meters. Stratigraphic subdivision follows the Gharif Thematic Scheme (Gharif Thematic Study; Alsop, 2015), with tops honored from adjacent fields and extrapolated into Field X.

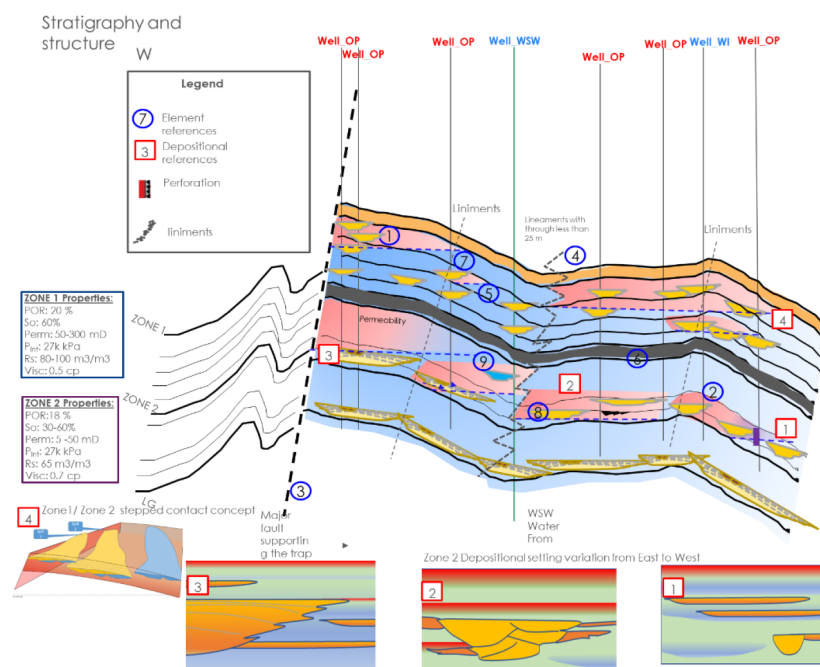


Figure 1—Stratigraphic cross-section (plumping diagram) showing the target reservoirs and wells penetrations.

Sedimentary Setting

According to the GTS framework, sediment supply to Zones 1 and 2 in the greater area originated from structural highs to the east, along the margins of the Arabian or Indian subcontinent. Zone 1 was deposited on a humid to tropical fluvial plain, with meandering, multistorey channel systems incising into flood plain siltstones and mudstones. Sedimentary structures include levee deposits, crevasse splays, intra-channel bars, and small deltaic lakes. Borehole imaging (FMI) data from Field X suggest a channel orientation of northeast-southwest and east-west, consistent with regional observations.

Zone 2 represents deposition in a continental to marginal marine setting. Its basal unit is characterized by tidally influenced shoreface sands, while the overlying units consist of fluvial channels and sheetflood sands developed across a semi-arid coastal plain. The main channel trends for Zone 2 are northwest-southeast, and marine influence increases westward across the regional stratigraphic trend.

Both zones are highly heterogeneous and laterally discontinuous. Despite 29 well penetrations in the southern area, delineating channel geometries and continuity remains challenging. Three cores confirm that Zone 1 consists of isolated and stacked fluvial channels interbedded with proximal and distal floodplain deposits. Additional core data from nearby fields were also incorporated.

The channel systems of both zones are generally oriented obliquely or perpendicularly to the main structural axis of Field X. This configuration provides potential for stratigraphic trapping in the north and south, while structural closures dominate to the east and west. Effective top seals are provided by intra-reservoir paleosols and lacustrine mudstones, while the regional Khuff Formation forms the upper seal. In some areas, erosional amalgamation between younger and older sands may reduce seal integrity.

Petroleum System

Field X serves as a benchmark for stratigraphic trapping in central Oman. Its structural geometry and stratigraphic configuration—characterized by isolated fluvial channel sands encased in mudstones—provide a reliable analog for hydrocarbon accumulation models. Core data from Zone 2 (primarily from its upper subunit) confirm a fluvial depositional environment with distal floodplain signatures.

Three principal depositional elements in Zone 2 were identified:

- Fluvial channels
- Multi-storey channels
- Paleosol/distal floodplain facies

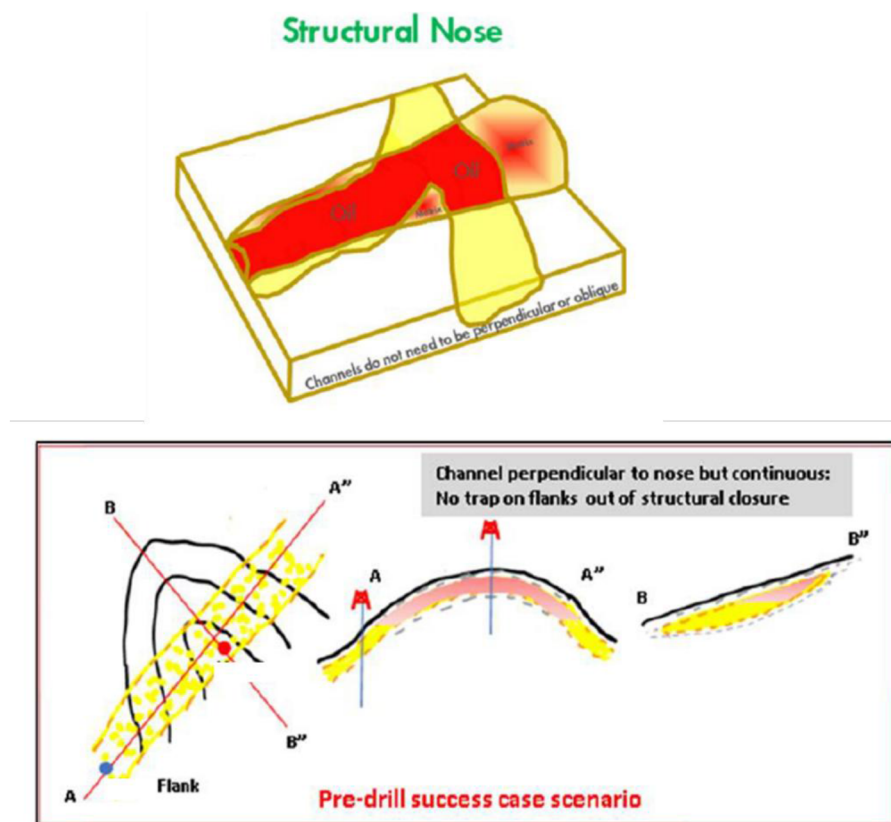


Figure 2—the structural setting and the reservoir orientation across the structure. The trapping Mechanism was consisting of Structural and Stratigraphic elements.

Depositional Model Summary

Zone 1 is modeled as a system of distributary and multi-storey fluvial channels with interbedded floodplain mudstones, paleosols, and lacustrine deposits. Channel geometries were varied in low, base, and high-case scenarios to assess development uncertainty.

Zone 2 is Characterized by braided fluvial channels influenced by tidal dynamics, interbedded with floodplain mudstones and paleosols. Variations in channel scale and tidal energy were incorporated across scenarios due to their impact on net-to-gross and localized diagenesis (e.g., calcite cementation).

Heterolithic Facies and Implications

Heterolithics—alternating sand and mud layers—are prevalent in both zones and play a significant role in controlling reservoir performance. In Zone 1, heterolithics occur within small fluvial channels, crevasse splays, and proximal floodplain environments. These deposits act as vertical baffles and barriers, and are associated with fining-upward log signatures and occasional ripple structures indicating channel abandonment.

In Zone 2, heterolithics correspond to distal floodplain and sheetflood environments. These contribute to increased vertical compartmentalization and reduced connectivity. Statistical analysis suggests heterolithics comprise 7–10% of Zone 1 and 20–30% of Zone 2, significantly impacting effective reservoir continuity and sweep efficiency under waterflood operations.

In summary, the depositional complexity and facies architecture in Field X present significant development challenges, with reservoir quality and continuity tightly controlled by channel distribution, facies stacking, and heterolithic content. Successful development depends on understanding and integrating these geological characteristics into modeling and planning workflows.

Methodology

This section outlines the techniques and tools applied to characterize the reservoirs, define fluid contacts, analyze equilibrium behavior, and assess the implications of contact tilt on field development. A multi-disciplinary approach combining petrophysical data, structural interpretation, capillary pressure modeling, and equilibrium time estimates was employed. In Field X case the field had not enough time to restore equilibrium due the rock properties in the tighter zone.

Tilted Contact Concept

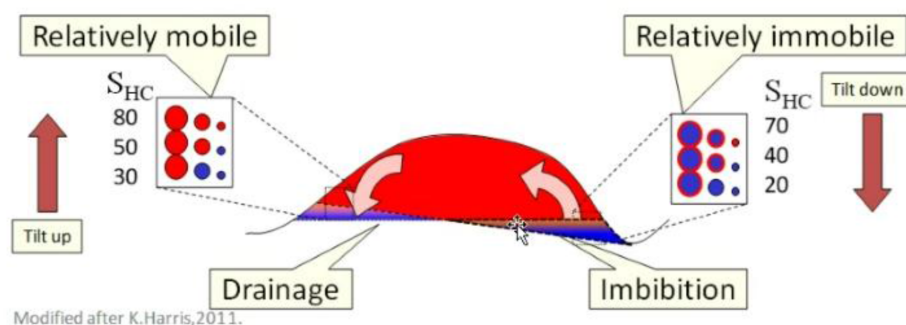


Figure 3—Conceptual model of a tilted contact caused by post-charge uplift. Oil drains on the uplifted side, while water imbibes on the downdip side, preserving the tilt due to mobility differences and slow equilibration. (Modified after K. Harris, 2011)

Reservoir Characterization

Reservoir properties were assessed using conventional core analysis, wireline logs, and dynamic production data. The upper reservoir shows consistent high porosity (~24–28%) and permeability (hundreds of mD), while the lower reservoir is tighter, with porosity averaging 10–14% and permeability below 10 mD. Facies mapping revealed that the upper reservoir is laterally discontinuous and structurally bounded, whereas the lower reservoir is more laterally extensive and stratigraphically trapped.

Formation tops and fluid contacts were picked across more than 100 wells. Log-based saturation analysis using Archie's equation and capillary pressure modeling confirmed the presence of varying saturation profiles, with some profiles indicating imbibition behavior rather than typical drainage curves—especially in the lower reservoir.

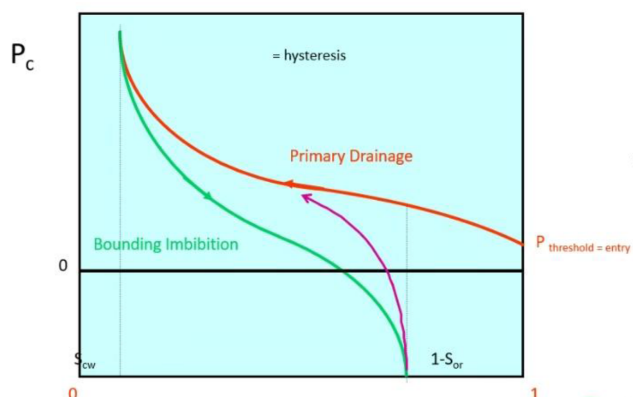


Figure 4—Capillary pressure curves showing primary drainage and bounding imbibition with associated hysteresis. The difference reflects slower equilibration during imbibition, relevant for interpreting fluid behavior in tilted contact scenarios. (Smith, J., and Doe, A., (2017)

Contact Mapping Techniques

Oil-down-to (ODT) and water-up-to (WUT) data were compiled to define the limits of mobile hydrocarbon presence. These contact markers were plotted in map view and cross-sections. The distribution of these data points indicated a tilted contact, trending from higher elevations in the north to lower levels in the south, following the axis of the structural nose.

The observed gradient could not be explained solely by capillary or height functions under equilibrium, suggesting a post-charge structural tilting event that disturbed the originally horizontal contact. In current development practices, shale barriers are used to define stepped contact zones between reservoir segments. However, this simplification may overlook continuous oil columns affected by structural tilting.

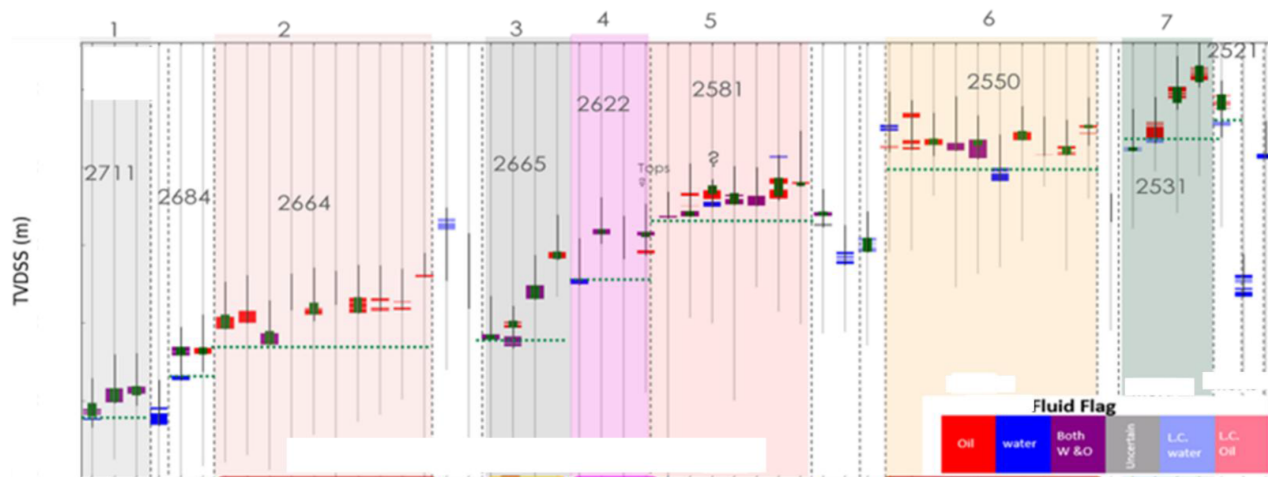


Figure 5—ODT and WUT Analysis of the field from North to south showing the apparent tilt of the contact following the dip of the structure

Equilibrium Time Estimation

To understand the persistence of tilted contacts, equilibrium time scales were estimated using diffusion-driven and capillary pressure-controlled equilibration models. The difference in behavior between drainage and imbibition curves is critical: the upper reservoir achieves equilibrium more rapidly due to higher permeability and favorable relative mobility, whereas the tight reservoir's capillary-driven imbibition process is much slower.

Simulations based on field-specific capillary pressure and mobility ratios estimate equilibrium times up to 20 million years in the lower reservoir. This supports the hypothesis that the observed tilted contact has been preserved due to slow equilibration in the tighter lithofacies.

Structural and Tectonic Analysis

Structural analysis of seismic and well data indicates a gentle anticlinal nose trending north-south across the field, with fault interpretation revealing limited compartmentalization. However, evidence of post-charge structural tilting is present in the evolution of this nose, suggesting that tectonic uplift occurred after hydrocarbon migration. To better understand the tectonic history and fluid distribution, selected shallow horizons were interpreted in seismic to provide structural control and insight into the regional tectonic framework. These horizons were strategically chosen to reconstruct the paleo-structure of the reservoir interval.

Through dip restoration of the reservoir surfaces, the pre-tilt structural configuration was reconstructed. This paleo-structure revealed a more symmetrical geometry with a flat fluid contact, significantly different from the present-day tilted configuration. Interestingly, the restored paleo-highs were located differently

compared to the current structural highs, supporting the hypothesis that hydrocarbon accumulation occurred prior to the final tilting event.

This reconstruction was essential in explaining the observed saturation distribution across the field, which deviated from predictions based solely on Saturation Height Functions (SHF). In particular, downdip wells exhibited higher-than-expected hydrocarbon saturations, which aligned with the paleo-structural trapping and fluid distribution before tilt-induced re-equilibration. These insights underscore the value of integrating structural restoration and paleo-structure mapping in refining contact interpretations and saturation models.

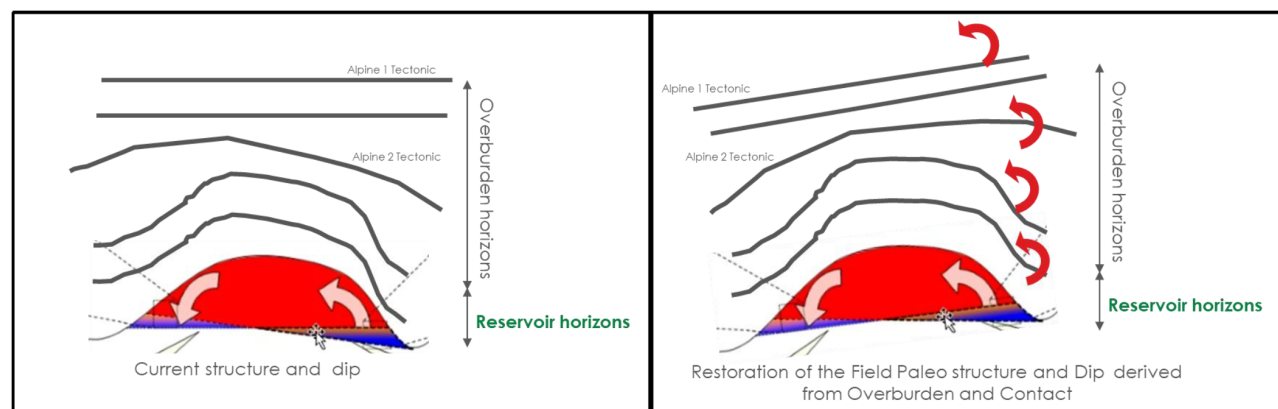


Figure 6—Comparison between the current structural configuration (left) and the restored paleo-structure (right) based on overburden and contact reconstruction. Restoration reveals a previously flat contact and different structural highs, explaining saturation anomalies and supporting the tilted contact concept.

Results and Observations

Contact Behavior and Tilt Effect

The tilted contact manifests as a 30–40 m elevation difference between the north and south of the field. This cannot be reconciled through simple structural dip, pressure depletion, or production effects, as it is observed in both dynamic and static wells. The stepping contact currently used for development does not align with the physical continuity of saturation gradients observed across the field.

The field-wide saturation distribution and transition zones provide further evidence of a tilted contact rather than a series of discrete hydraulic compartments.

Mobility and Capillary Pressure Curves

Laboratory measurements and SCAL data show marked differences between the drainage and imbibition capillary pressure curves in the two reservoirs. The upper reservoir displays steep drainage curves with minimal hysteresis, while the tight reservoir exhibits significant hysteresis and flatter curves under imbibition.

Relative permeability data confirm that oil mobility is high in the uplifted (drainage-dominated) areas, while water slowly invades the oil zones in the southern limb (imbibition-dominated). This leads to an asymmetric fluid distribution—stable in the tight reservoir due to low permeability and slow displacement.

Implications for Hydrocarbon Distribution

Traditional volumetric estimates assuming horizontal or stepped contacts may underestimate oil columns in the south and overstate them in the north. By adopting the tilted contact as a structural framework, additional oil volumes could be recognized—especially in areas currently interpreted as water-bearing but falling within the tilted transition zone.

This has direct implications on infill drilling opportunities and bypassed pay targeting. Mapping the contact as a continuous tilted surface rather than discrete stepped segments provides a better fit to the saturation data and supports more efficient development planning.

Discussion

The findings from this study highlight the importance of integrating geological, petrophysical, and tectonic insights when interpreting fluid contacts in complex clastic systems. The presence of an apparent tilted contact across both reservoirs challenges the assumption of capillary equilibrium in static models, particularly when dealing with stratigraphically trapped and low-permeability formations.

Drainage vs. Imbibition Dynamics

The asymmetric contact behavior observed between the two reservoirs is explained by the difference in drainage and imbibition processes. In the structurally uplifted areas, oil re-migrated and displaced water under a drainage regime, restoring partial equilibrium. In contrast, the downdip reservoir portions experienced imbibition, where water gradually invaded the oil-filled pore spaces. Due to capillary pressure hysteresis and low permeability, this process progressed slowly and incompletely, preserving a tilted contact.

This dual-mode displacement dynamic is a key observation, as it explains why the equilibrium state has not been fully restored over millions of years. It also provides a predictive framework for identifying similar tilted systems in other fields, particularly in post-charge tectonic settings.

Stepping vs. Tilted Contact Approaches

The current field development plan applies a stepping contact model, assuming localized barriers segment the reservoirs into discrete compartments. While this approach simplifies volumetric and development planning, it may lead to misinterpretation of saturation data and underestimation of connected pay.

In contrast, the tilted contact model, though more geologically complex, aligns better with the saturation transition observed in log data and capillary pressure trends. This model allows for a more continuous representation of the hydrocarbon column and could enable better placement of future wells, especially in downdip areas previously deemed water-wet.

Adopting this model may also help reduce dry hole risk and improve resource classification, particularly in cases where saturation uncertainty is high.

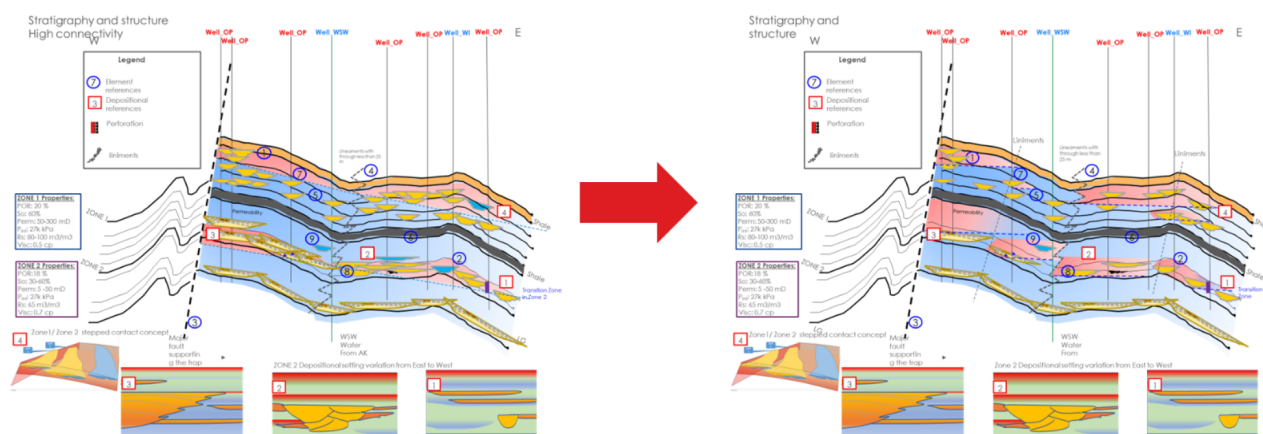


Figure 7—Stratigraphic cross-section (plumping diagram) before (left) and after (right) reinterpretation of structural dip and contact alignment. The updated model incorporates tilted contact understanding and paleo-structure reconstruction, leading to revised fluid distribution, improved correlation of depositional elements, and reclassification of potential pay zones into a stepping contact.

Development Strategy Optimization

Adopting a tilted contact interpretation requires updates to mapping, volumetric calculations, and well planning. Rather than assuming flat or stepped contacts, surfaces should be redefined as inclined planes that reflect the structural tilt. This improves the geological consistency of the model and better explains observed saturation variations.

Marginal or downdip zones previously considered non-pay may hold bypassed hydrocarbons. Re-evaluating these areas using saturation-height functions and SCAL data can unlock additional volumes and optimize well placement. Recovery factors and STOIP estimates should also be revised to reflect the increased hydrocarbon column height under the tilted model.

This approach enhances infill targeting, improves volumetric accuracy, and opens new development opportunities. It also highlights the need for additional SCAL, pressure, and saturation data to further refine the contact model and reduce subsurface uncertainty.

Conclusions and Recommendations

This study demonstrates that in clastic reservoirs where tectonic uplift occurs after hydrocarbon charge, tilted fluid contacts can be preserved—particularly in low-permeability settings where capillary-driven equilibration is inherently slow. Through the analysis of over 100 wells across two contrasting reservoirs, it was observed that an apparent tilted contact, trending from north to south, consistently aligns with the structural nose of the field. This contact tilt cannot be fully explained by standard capillary equilibrium assumptions alone, pointing instead to a post-charge tectonic disturbance that modified the original fluid distribution.

The upper reservoir, characterized by high rock quality, has shown signs of partial re-equilibration following structural tilt. In contrast, the tighter, lower reservoir retains a disequilibrium state that reflects its slower imbibition-dominated response. This asymmetric fluid behavior underscores the importance of understanding drainage versus imbibition processes when interpreting contact positions and saturation distributions in heterogeneous systems.

While the current development approach employs a stepping contact model for operational simplicity, this method may obscure the continuity of hydrocarbon-bearing zones and under-represent the true extent of the hydrocarbon column. By adopting a tilted contact interpretation, supported by structural restoration and saturation profiling, a more geologically consistent and volumetrically accurate model can be achieved.

Based on these findings, several recommendations are proposed. First, contact interpretations in stratigraphically trapped clastic reservoirs should be re-evaluated where there is evidence of post-charge tectonic deformation. Second, capillary pressure data and saturation-height modeling should be incorporated more rigorously into contact mapping workflows, particularly in low-permeability zones. Third, development strategies should be adjusted to incorporate insights from tilted contact models, as these may highlight untapped opportunities in downdip or structurally redefined areas.

Future work should focus on 3D simulations of fluid migration in tilted systems to better capture dynamic behavior under post-charge structural changes. Additional SCAL data from downdip wells will help constrain saturation functions, while probabilistic volumetric assessments should be used to quantify resource uncertainty under different contact assumptions. Together, these efforts will enhance field development planning and reduce risk in geologically complex clastic settings.

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